A1.2 Nucleic acids

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-a1-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 25–33 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Structure

A1.2.11 Direction in DNA and RNA

pp. 25–26

Phosphodiester bonds link carbon 3′ of one sugar to carbon 5′ of the next, so each strand has a **5′ end** (phosphate) and a **3′ end** (free -OH). The two strands of DNA are **antiparallel**. New nucleotides join only the 3′ end, so DNA and RNA are always built 5′ to 3′. In replication, the two strands are copied in opposite directions; in transcription, RNA polymerase reads the template strand 3′ to 5′; in translation, ribosomes read mRNA from 5′ to 3′.

A1.2.12 Purine-pyrimidine pairs

pp. 26–27

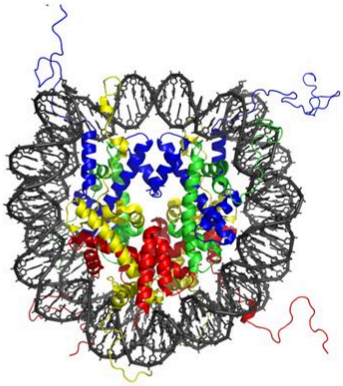
Purines (adenine, guanine) have two rings; **pyrimidines** (cytosine, thymine) have one. Each base pair is one of each, so A-T and C-G pairs are the same width and the helix keeps the same shape whatever the sequence. Hydrogen bonds (two in A-T, three in C-G) hold the pairs together.

A1.2.13 Nucleosomes

pp. 27–29

The nucleosome

A1.2.13



Eukaryotic DNA winds about twice around a core of **eight histones** (two each of four types): a **nucleosome**.

Linker DNA joins nucleosomes; histone **H1** holds the DNA where it enters and leaves.

Positive histones (lysine, arginine) attract negative DNA phosphates.

Nucleosome (1KX5): DNA in grey around histones. Zephyris, CC BY-SA 3.0.

Like beads on a string. Prokaryotic DNA has no histones: it is "naked".

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In eukaryotes, DNA is wound about twice around a core of **eight histone proteins**. **Linker DNA** runs to the next nucleosome, and an **H1 histone** holds the DNA in place. Positively charged lysine and arginine in histones attract the negative phosphates of DNA. Packing fits about 2 m of DNA into each nucleus, lets chromosomes condense for division, and helps control which genes can be transcribed.

Evidence

A1.2.14 Hershey and Chase

pp. 29–30

In 1952, phages were labelled with ^{35}S (in protein) or ^{32}P (in DNA) and allowed to infect *E. coli*. Blending removed the phage coats and centrifuging separated them from the cells. Most ^{32}P went into the bacteria and some appeared in new phages; most ^{35}S stayed outside. So DNA, not protein, carries the genetic information.

A1.2.15 Chargaff's data

pp. 31–32

Chargaff measured base proportions in DNA from many species. In every species **A = T and G = C**, so purines equal pyrimidines; but the proportions differ between species. This **falsified** the tetranucleotide hypothesis, which said DNA was a simple repeat of the four bases and could not carry information, and it supported complementary base pairing.

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